SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-25 00:39 UTC

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Generated 2026-04-25 00:39 UTC

- Cycles recorded: **50**
- Time span: 27.0h (~1.85 cycles/h)
- Notebook: `/home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl`

Verdict distribution

Verdict	Count
INCONCLUSIVE	39
FALSIFIED	8
SUPPORTED	3

Mathematical fields explored (field_A)

Field	Cycles
Noncommutative geometry	2
Combinatorial homotopy theory	1
Kazhdan-Lusztig theory of Hecke algebras	1
Combinatorial algebraic topology	1
Knot theory (quantum invariants)	1
Quadratic forms over finite fields (Grothendieck-Witt groups)	1
Clifford algebras over finite fields	1
Toric geometry (via Gröbner degenerations)	1
Arithmetic geometry (Tate-Shafarevich groups)	1
Geometric Langlands correspondence	1
Algebraic graph theory (chromatic polynomials)	1
Clifford algebras over real vector spaces	1
Algebraic cycles (Chow groups over finite fields)	1
Algebraic lattice theory (Möbius functions of flow lattices)	1
Directed algebraic topology (d-space homology)	1
Motivic integration (Denef-Loeser zeta functions)	1
Matroid theory (Clifford index of binary matroids)	1
Representation theory of finite groups (Frobenius-Schur indicator)	1

Field	Cycles
Categorification and knot Floer homology	1
Modular forms	1
Algebraic geometry (ideals in polynomial rings)	1
Hypergraph Tutte polynomials	1
Galois theory of finite field extensions	1
Combinatorial homotopy theory (unstable 2-cohomotopy)	1
Geometric group theory (Kazhdan's property T)	1
Quadratic forms over GF(2)	1
Commutative algebra (Castelnuovo-Mumford regularity)	1
Algebraic K-theory (Bass Nil-groups)	1
Arithmetic geometry (Selmer groups of elliptic curves)	1
Toric geometry (via initial ideals and monomial degenerations)	1

Last 15 cycles (chronological)

Time	Verdict	Title
2026-04-24 16:30 UTC	`INCONCLUSIVE`	Vandermonde Rank of Clause-Variable Incidence Matrix Predicts Res
2026-04-24 17:19 UTC	`INCONCLUSIVE`	Average Sensitivity of Clause-Indicator Function Bounds Resolutio
2026-04-24 18:09 UTC	`INCONCLUSIVE`	Minimal ABP Size and Permutation Polynomial Degree
2026-04-24 18:43 UTC	`INCONCLUSIVE`	Hypergraph Discrepancy and Communication Complexity of 3-SAT
2026-04-24 19:18 UTC	`INCONCLUSIVE`	Hilbert Function of Ideal Bounds Communication Complexity
2026-04-24 19:54 UTC	`INCONCLUSIVE`	Matrix Rigidity Bounds Communication Complexity of 3-SAT Instance
2026-04-24 20:34 UTC	`INCONCLUSIVE`	Ideal Generator Count and Bounded Arithmetic Proof Size
2026-04-24 21:05 UTC	`INCONCLUSIVE`	Orbit Signature Decay Implies Communication Entropy Barrier Growt
2026-04-24 21:22 UTC	`INCONCLUSIVE`	ACC^0 Circuit Size Bounded by Finite Field Equation Solution Coun
2026-04-24 22:09 UTC	`FALSIFIED`	Average-Case SAT Hardness Linked to Finite Field Solution Counts
2026-04-24 22:27 UTC	`INCONCLUSIVE`	Specht Module Dimensions Bound ABP Size for Permutation Polynomia

Time	Verdict	Title
2026-04-24 23:06 UTC	`INCONCLUSIVE`	Seed Length of Nisan-Wigderson PRG from Finite Semifields
2026-04-24 23:33 UTC	`INCONCLUSIVE`	Tseitin Resolution Width and Expander Eigenvalues
2026-04-25 00:09 UTC	`INCONCLUSIVE`	Finite Geometry Line Count Bounds EF Proof Size
2026-04-25 00:39 UTC	`INCONCLUSIVE`	Dimension of Variety from Tautology Ideal Bounds EF Proof Size

How to read the reports

- `reports/supported_findings.md` — SUPPORTED conjectures with full test code. Paper-quality.
- `reports/falsified_findings.md` — FALSIFIED with counterexamples.
- `reports/daily_YYYY-MM-DD.md` — chronological log by day.
- `pvsnp_notebook.jsonl` — raw JSONL, one entry per cycle (source of truth).

PDF export

Install `pandoc` + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```